EC Declaration of Conformity
EC Guideline Electro Magnetic Compatibility 2004/108/EC
Machinery Directive 2006/42/EC has been fulfilled.

The safety targets regarding the Low Voltage Directive 2006/95/EC according to the following co-ordinated standards have been applied:

BSEN 60335-1:2002 +A14:2010 – Refrigeration, Household Safety
BSEN 60335-2-89:2003 – Refrigeration, Motors and Pumps

Household and Similar Electrical Appliances – Safety (particular requirements for commercial refrigerating appliances with an incorporated or remote refrigerant condensing unit or compressor)

BSEN61000-6-3:2001 Electromagnetic emissions compatibility (EMC)- Generic standards.

BSEN61000-6-1:2001 Radiated and Immunity compatibility (EMC) – Generic standards.

CE

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Multiplex

icecore eco-friendly

ICECORE 1 Soft Drink Postmix (Cold Carbonated)

Hydrocarbon Shelf Cooler

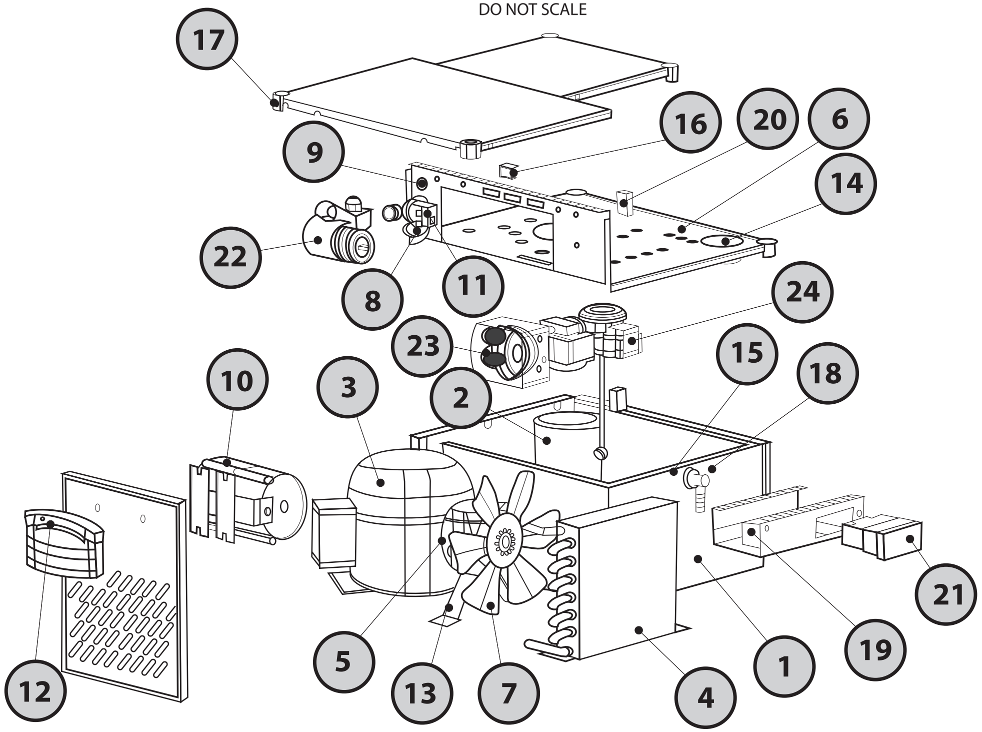
USER, INSTALLATION & MAINTENANCE MANUAL

PRODUCT CODE	02-3000-XX
MANUAL PART No.	PI56788
MANUAL REVISION:	Rev 3

DATE: 12/04/12

Manitowoc

DO NOT SCALE



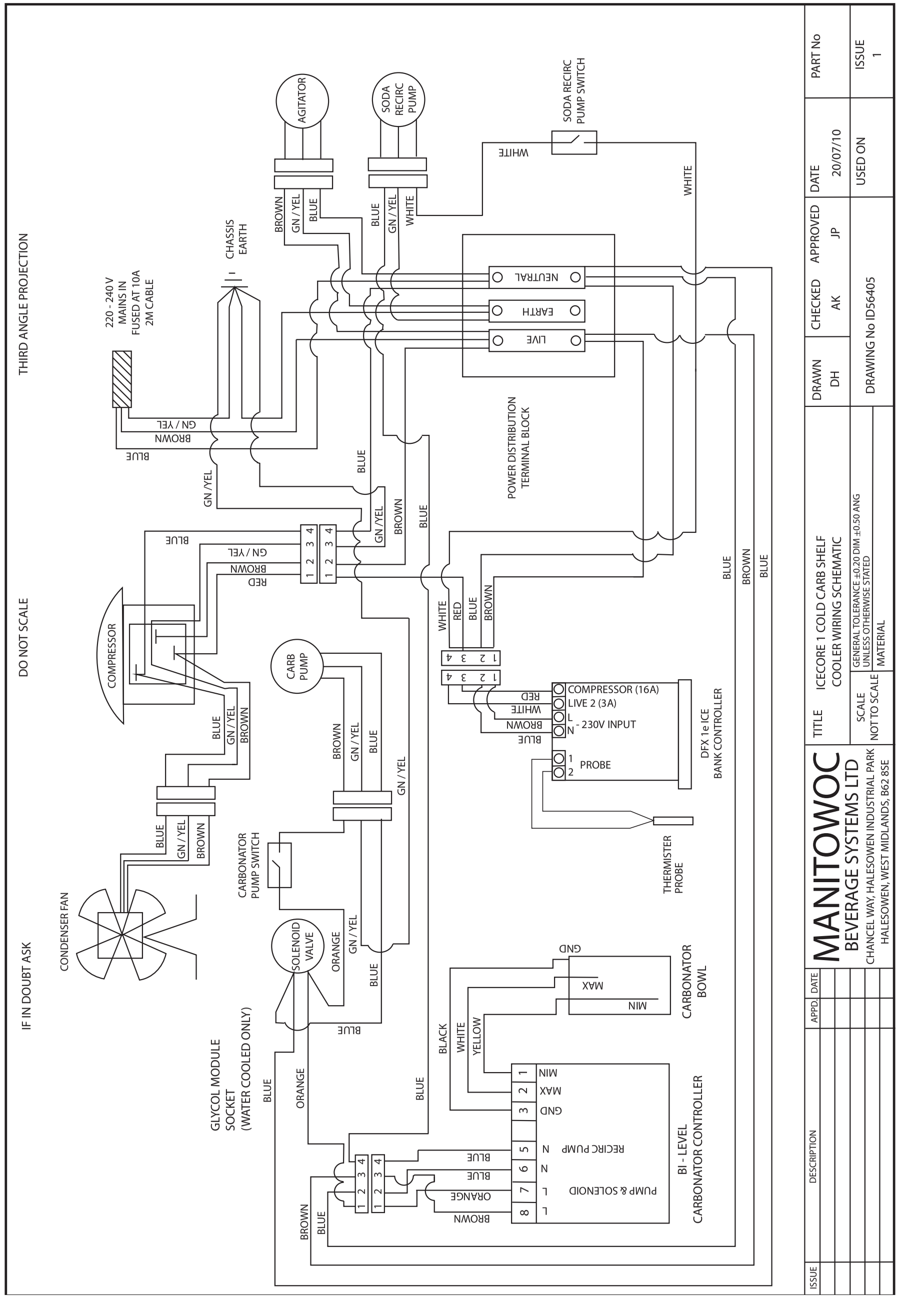
IF IN DOUBT ASK

ITEM	PART No	DESCRIPTION	QTY
1	BA45137	OPTIMA 7 WATER BATH	1
2	CB56100	CARBONATOR BOWL	1
3	CM58083	TL5CN COMPRESSOR	1
4	CN55546	COPPER/ALUMINIUM CONDENSER	1
5	MO48894	EBM 9W UL FAN MOTOR	1
6	OP73636	8mm (5/16”) BLACK SNAP BUSH	10
7	OP73807	FAN BLADE	1
8	OP73838	3/8” X 3/8” BULKHEAD FITTING	1
9	OP73972	IMM MAINS CABLE STRAIN RELIEF	1
10	PI42126	RPM MOTOR	1
11	PI47525	WATER SOLENOID VALVE JG 3/8 ENDS	1
12	PI47648	MOULDED PLASTIC LIFTING HANDLE	1
13	PI48892	FAN BRKT 58 HIGH EBM	1
14	PI49067	FILLER CAP	1
15	PI49133	GROMMET	1
16	PI49187	RIVETABLE EARTH STUD	1
17	PI56129	LID CORNER MOULDING	6
18	PI56179	LONG OVERFLOW WITH VENT	1
19	PI56242	BI-LEVEL CARB CONTROL MK 2	1
20	PI56363	SLIDE SWITCH	2
21	PI56613	DFX1E C/W SHORT PROBE	1
22	PU70172-01	PROCON BRASS PUMP	1
23	SA53096	SODA RECIRC PUMP	1
24	SA53097	AGITATOR MOTOR	1

NOT SHOWN	PI56643	COMPRESSOR PTC START DEVICE	1

CORE SPARE PARTS FOR MODEL 02-3000-01

WIRING DIAGRAM FOR MODEL 02-3000-01



GENERAL DESCRIPTION, FUNCTION AND INTENDED USE OF THIS EQUIPMENT.

- This equipment is a hermetically sealed refrigeration unit in the form of a shelf cooler.

- This equipment is designed to be installed under bars & counters and carbonate, recirculate and chill mains potable water and chill soft drinks syrup.

- This equipment should be installed in the system between the bulk storage area and the drinks dispense font(s).

- This equipment provides cooling to the water and up to 4 syrup products by means of cooling coils within a water filled tank in which an ice bank will form when the unit is switched on. This ice bank size is controlled by the electronic controller and is factory pre-set.

- This equipment has an integral carbonator which carbonates mains potable water using CO2. The carbonated water is recirculated through cooling coils from the carbonator via an internal recirculation pump through a python to a dispense font(s).

- This equipment uses hydrocarbon refrigerant (R290), and having intelligent ice bank water bath control, the ICECORE-1 is capable of delivering substantial energy savings over conventional shelf coolers, whilst providing the same level of service.

- There are **NO** other recommended uses for this equipment.

WAYS THE EQUIPMENT SHOULD NOT BE USED

- This equipment should be used only in accordance with the instructions above. It should not be used for any other purpose.

- Do not use the product coils to chill other products than those stated above.

- Do not operate the equipment in a wet environment, any spillage must be wiped dry immediately.

- The unit should not be installed in small enclosed spaces such as cupboards or pantries, where fresh air flow is restricted.

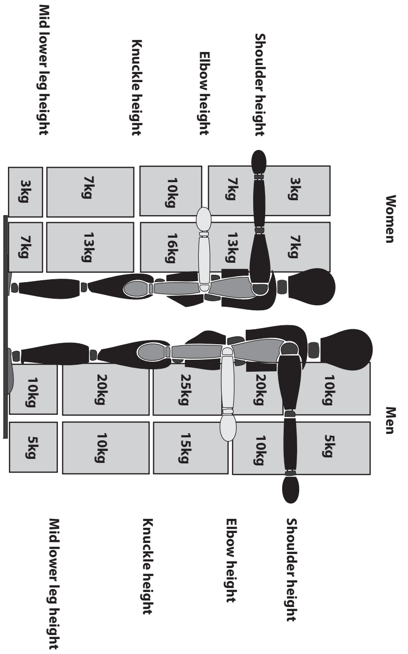
- Keep the unit free from excessive heat and cold. Minimum and maximum ambient temperatures are:
  Minimum: 5°C
  Maximum: 32°C

- Misuse or use of the equipment for any other purpose than those identified above will invalidate any warranty, and may constitute a danger to yourself and others.

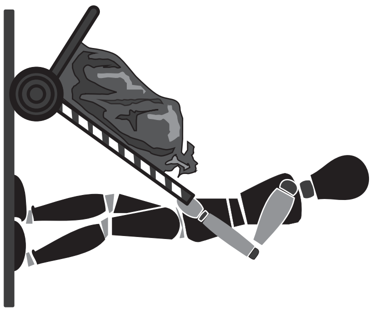
GENERAL DESCRIPTION, FUNCTION AND INTENDED USE OF THIS EQUIPMENT

This equipment can weigh up to 25kg dependant on unit specification. Please refer to the **recommended** safe handling guide below before moving this equipment.

- Before lifting or moving this equipment it is recommended that all persons performing these tasks should receive relevant training in safe handling.



- All persons lifting or moving this equipment must be wearing the correct personal protective equipment.
- Lifting handles on the side panels are provided to assist in the safe lifting and carrying of this unit. Avoid lifting the unit from the base.
- It is recommended that the cooler unit should be lifted or carried by 2 people.
- To prevent personal injury, where practical, transporting of the unit over extended distances should be done using a mechanical aid.
- Lifting the unit above knuckle height is not recommended, except with the assistance of mechanical or hydraulic lifting equipment such as a hoist. Any lifting equipment used must be rated to support the weight of the unit.
- Never lift or move the unit when it is filled with water.
- Always transport the unit in the correct upright position, never on it's side or top.



Consider mechanical aids; even a sack truck can make a big improvement.

GROUP	TYPE	SPEC	INDEX	TECHNICAL SPECIFICATION SHEET	ISSUE No 2
02	3	000	01	**ICECORE 1 COLD CARB SHELF COOLER**	16/08/2011

	PHYSICAL DATA		**PUMPS / AGITATORS / MOTORS**
Overall Dimensions	W= 605mm D= 401mm H = 285mm	**Soda Recirc Pump**	ITT/TOTTON - HBR 6/8 RE-Gen pump
Operating Weight	40.9 Kgs	**Agitator**	3/4" Stack 110mm shaft shaded pole motor
Dry Weight	30.5 Kgs	**Condenser Fan Motor**	9 Watt EBM induction motor
Water Bath Capacity	10.4 Litres	**Condenser Fan**	8" x 40˚ pitch aluminium
	REFRIGERATION SYSTEM	**Carbonator Pump**	Fluid-O-Tech 100Gall/Hr brass positive displacement
Compressor	TL5CN COMPRESSOR (R-290)		**PRODUCT COILS**
Duty	345 Watts @ - 10˚C EVAP TEMP	**Syrup**	3 x 5/16" Coils, 1.5 meters AV. length
Evaporator	3/8" x 10 meters	**Pre-chill**	1 x 5/16" Coil, 7.5 meters long
Condenser	7 Row 2 bank copper alluminium	**Soda Coil**	1 x 5/16" Coil, 7.5 meters long
Refrigerant Flow Control	0.047" Capillary x 3.8 meters	**Still Coil**	1 x 5/16" Coil, 7.5 meters long
Refrigerant	75 Grams R-290		**CONSTRUCTION**
	CONTROLS	**Outer Panels & Lid**	Mild steel plastic coated
Ice Bank Control	DFX series 1e electronic control (thermister)		Alluminium/Zinc alloy & esetic tex paint
Panel Switches	2 Slide switches for soda pump & carb pump isolation	**Base Plate**	Galvanised mild steel
Indicators	Electronic display on ice bank control		**PERFORMANCE**
Carbonator Control	Electronic resistance, BI-level probes + GND	**Ice Bank**	3.8 kgs (Nominal)
	SYSTEM PROTECTION	**Max Air on Temperature**	32˚C
Compressor	Thermal self reseting switch on compressor	**Carb Fill Rate**	1.5 OZ/SEC
Carbonated Pump	Time out of carb pump after 1 min 30 secs if bowl does not fill	**Dispense Rate**	58x10 0z drinks below 4.4˚C at 2x10 0z/min
Soda Recirc Pump	Time out of soda recirc pump after 1 min 30 secs if bowl does not fill		(Test conducted via 12M python in 24˚C ambient using
Mains fuse	10 AMP fuse in plug		24˚C water & syrup in a 5.1 ratio)
	ELECTRICAL	**Ice Bank Pulldown Time**	1hr 42 mins +/15mins (No Python)
Run Current	4.09 Amps	**Recovery rate (0˚C- Full ice)**	2hrs 10 mins +/-5 mins (12M Python in 24 ˚C ambient)
Voltage	220-240 Volts AC		
Power Consumption	821 Watts		
Frequency	50 Hz		
Mains Lead/Connector	3 Pin UK mains lead fused at 10A		
	WATER AND CO2 PRESSURES		
Max/Min Water Pressure	50/20 psi (3.4 /1.36 bar)		
Max/Min Co2 Pressure	80/75 psi (5.44 /5.1 bar)		

TECHNICAL SPECIFICATION FOR MODEL 02-3000-01

FOR QUALIFIED ENGINEERS ONLY

METHOD OF POWER ISOLATION

1. Switch off the unit at the plug socket.
2. Remove the plug from the socket.

DISMANTLING / DECOMMISSIONING PROCEDURE

1. Isolate the unit from Mains Supply.
2. Turn off the water supply.
3. Drain the system down using the CO2 pressure to empty the carbonator bowl, soda recirculation lines and syrup lines.
4. Turn off the CO2 supply at the bottle and de-pressurise the system.
5. Disconnect the unit from the python product lines and the water and CO2 supply.
6. Siphon or pump out the water from the bath.

IMPORTANT: FAILURE TO REMOVE ALL COOLANT COULD RESULT IN SUBSTANTIAL AMOUNTS OF COOLANT BEING RELEASED FROM THE UNIT WHICH MAY BE DETRIMENTAL TO THE UNIT AND/OR ITS SURROUNDINGS. AND ALSO INCREASE THE MAXIMUM WEIGHT OF THE UNIT IF NOT REMOVED.

PREVENTION OF FREEZING / ACTION REQUIRED IF FREEZING OCCURS.

This unit must not be sited in such a way as not to expose it to temperatures likely to cause freezing i.e. below 0°C.

If the Unit is to be sited in an unheated area then it would be wise to insulate all pipe work and provide some form of Emergency Heating which should be controlled by a Frost Stat, and sited in the close proximity to the Unit.
In the event of Freezing Up occurring, the following action is recommended:-

1. Isolate the unit from Mains Electricity Supply.
2. Isolate Product Supply from unit and de-pressurise the system.
3. Apply gentle warmth to the general area of the unit and its Pipe work.
4. Check for obvious leaks.
5. Turn product supply on / re-pressurise the system whilst continually watching for leaks.
6. Reconnect Mains Electricity supply

Observe the unit running for a short period watching out for Leaks, Strange noises or any other form of malfunction, if no problems are observed then normal operation of the unit may be resumed.

WARNINGS AND SAFETY INFORMATION.

Warning

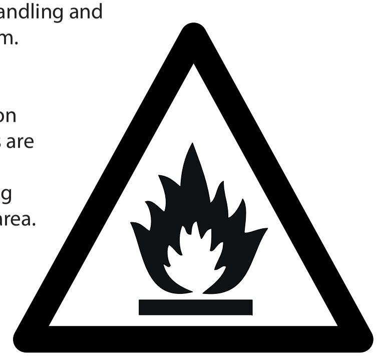
THIS EQUIPMENT IS CHARGED WITH R290 REFRIGERANT (PROPANE).

**ONLY QUALIFIED SERVICE ENGINEERS HOLDING A VALID HANDLING CERTIFICATE FOR CARE 40 (PROPANE) CAN WORK ON THE REFRIGERATION SYSTEM OF THIS EQUIPMENT.
PLEASE READ THE INFORMATION BELOW BEFORE ANY WORK IS CARRIED OUT.**

Refrigeration R290 (Care 40, Propane)

Note: Only engineers who have been trained in the safe handling and use of hydrocarbon refrigerants should work on this system.

- Work on this system in a well ventilated area or outside.
- Use a local leak detector to indicate if there is hydrocarbon in the air around the system (place it at a low level as HCs are heavier than air)
- Ensure there are no sources of ignition (flames or sparking electrical components) within 3m (10 feet) of your work area.
- If replacing components use like for like replacements.
- Take great care when brazing to ensure all HC has been removed from the system.



Use refrigerant grade propane (R290 or Care 40)

Warning THIS EQUIPMENT MUST BE EARTHED

Warning THIS EQUIPMENT SHOULD BE FITTED WITH A CORRECTLY FUSED AND WIRED 13 AMP PLUG FITTED WITH A 10 AMP FUSE OR ALTERNATIVELY A STANDARD EURO PLUG TO IEC83:1975.

Warning IF THE MAINS LEAD FITTED TO THIS EQUIPMENT IS DAMAGED IN ANY WAY IT SHOULD BE REPLACED. THIS WORK SHOULD ONLY BE CARRIED OUT BY A QUALIFIED ELECTRICAL ENGINEER.

Warning THE WIRES IN THE MAINS LEAD ARE COLOURED IN ACCORDANCE WITH THE FOLLOWING CODE:
Green and Yellow.......................................Earth
Blue...Neutral
Brown...Live

The unit is supplied with a sealed mains plug, fused to 10 amps. If this plug is removed for any reason, a replacement plug may have a different wiring colour protocol. Below is the detail for resolving the differences:

a) The wire which is coloured Green and Yellow MUST be connected to the terminal in the Plug that is marked with the letter, E™ or by the earth symbol ⏚ coloured Green or Green and Yellow.

b) The wire which is coloured Blue MUST be connected to the Terminal in the Plug that is marked with the letter, N™ or coloured Black.
c) The wire which is coloured Brown MUST be connected to the Terminal in the Plug that s marked with the letter, L™ or coloured Red.

All mains voltage components are hazardous. Any electrical work must be carried out by a qualified competent person.

⚠ Warning

THIS EQUIPMENT CAN CONTRIBUTE TO THE AMBIENT TEMPERATURE.

⚠ Warning

MAXIMUM AND MINIMUM OPERATIONAL AMBIENT TEMPERATURES.
MINIMUM 5 °C
MAXIMUM 32 °C

⚠ Warning

ALTHOUGH EVERY CARE IS TAKEN DURING MANUFACTURE, DAMAGE TO THE METALWORK DURING TRANSPORT, INSTALLATION AND GENERAL USE MAY OCCUR, RESULTING IN SHARP OR JAGGED EDGES. AVOID CONTACT WITH METAL EDGES OR OTHER POTENTIAL HAZARDS.

⚠ Warning

ALWAYS KEEP THE WATER BATH FILLER CAP FITTED.

⚠ Warning

DO NOT DROP ANY HARD MECHANICAL PART INTO THE WATER BATH – THIS MAY CAUSE PUMP FAILURE.

⚠ Warning

SOME COMPONENTS WILL ROTATE FOR A SHORT PERIOD AFTER THE POWER TO THE UNIT HAS BEEN SWITCHED OFF. THESE COMPONENTS SHOULD BE AVOIDED UNTIL STATIONARY.

⚠ Warning

THE EVAPORATION TEMPERATURE IN THE HERMETIC CIRCUIT CAN REACH AS LOW AS -10°C. WITHOUT TAKING PREVENTIVE STEPS THIS CAN BE A POTENTIAL SOURCE OF ACCIDENTS WHILST CLEANING OR MAINTENANCE OF PARTS AT SUCH LOW TEMPERATURES.

⚠ Warning

THE COMPRESSOR, CONDENSER, HIGH PRESSURE REFRIGERATION TUBES AND MOTORS WILL BECOME HOT DURING OPERATION. PLEASE AVOID ANY CONTACT TO THESE PARTS DURING AND AFTER OPERATION UNTIL THEY HAVE COOLED DOWN.

⚠ Warning

IN THE EVENT OF ACCIDENTAL OVER-FILLING OR CONTAMINATION BY DEBRIS, SEEK ADVICE FROM THE SERVICE PROVIDER WHO MAINTAINS AND SERVICES THIS EQUIPMENT. ACCIDENTAL OVER-FILLING COULD CAUSE WATER TO FLOOD INTO ELECTRICAL COMPONENTS, CREATING A SERIOUS HAZARD.

⚠ Warning

THE WATER COOLED VERSION OF THIS UNIT RECIRCULATES COOLING WATER THROUGH A HEAT EXCHANGER TO COOL THE REFRIGERANT GAS. THIS COOLANT WATER CAN REACH TEMPERATURES OF 50°C DURING PEAK PULLDOWN. CARE MUST BE TAKEN TO PREVENT SCALDS AND BURNS IF THE UNIT DEVELOPS A BURST COOLANT TUBE OR LEAK.

⚠ Warning

THE COOLING COILS OF THIS UNIT ARE UNDER PRESSURE DURING USE AND IT SHOULD BE NOTED THAT THE CARBONATION PROCESS, AND IN SOME CASES THE BLANKET PRESSURE TO DISPENSE BEERS, INVOLVES THE USE OF HIGH PRESSURES AND POTENTIALLY NOXIOUS GAS AND AS SUCH DUE CARE SHOULD BE TAKEN WHEN HANDLING, INSTALLING AND MAINTAINING THE EQUIPMENT WITH PARTICULAR REGARD TO THESE HAZARDS.

FOR QUALIFIED ENGINEERS ONLY

MAINTENANCE REQUIREMENTS

Monthly / Every visit: By a Competent Service / Maintenance Engineer.

1. Isolate unit from mains electricity supply.
2. Remove any extraneous debris from the unit or its casing preferably using a Vacuum cleaner or brush.
3. Clean the condenser (integral unit only)
4. Check the water bath level. Top up if required.
5. Clean the product lines (Frequency may be dependant on product being used).
6. Check unit for electrical safety.

RECOMMENDED LINE CLEANING METHOD

IMPORTANT: Persons performing Cleansing /Sanitizing operations MUST be competent and fully trained in safe methods in the use of Cleansing / Sanitizing Agents and their applications.

IMPORTANT: Personal Protective Equipment Should Always Be Used.

IMPORTANT: Do Not Use A Water Or Steam Hose To Clean The Unit Whilst Still Installed.

Clean and sanitize the cooling coils and product lines using a proprietary cleanser/sanitizer of the Alkaline Hypochlorite type, in accordance with the manufacturer's recommendations. MBS recommends the use of "Pipeline" beer line cleaner, which changes colour according to the condition of the product lines. It is available from beer and soft drinks wholesalers, or direct from:
Chemisphere UK Ltd, 143-149 Bath Road, Kettering, Northants NN16 8NE.
See above for detailed cleaning instructions.

Cleaning instructions for "Pipeline" beer line cleaner:

1. Draw clean water through the system until it all traces of product have disappeared.

2. Mix the beer line cleaner with clean water, in the ratio specified in the instructions [typically 1 capful per 4.5 litres (gallon)]. The solution will be purple in colour. NOTE: use the appropriate personal protective equipment, to avoid splash burns or damage to eyes from splashes.

3. Connect the line cleaning solution to the lines to be cleaned.

4. Draw the line cleaning solution through the dispense tap. If the lines are particularly contaminated with yeast or bacterial growth, the run-off will be green/grey/black.

5. Close the tap and leave for 10 minutes.

6. Repeat stages 4 and 5 twice, or until the run-off is purple in colour.

7. The purple colour denotes lines which are clean and free from contamination.

8. Disconnect the line cleaning solution and draw fresh water through the system. Dispense some water into a glass or other clear vessel and check that there is no purple colour remaining. If necessary, draw more clean water through the system until the run-off is clear.

9. The lines are now clean, safe and ready for reconnection to the product. Reconnect the Keg in the storage/cellar area and pull beer through to the tap.

3. To adjust the opening and closing hours:
Note: This feature allows up to 2 opening and closing sessions per day. The Manitowoc default setting assumes a single 12-hour operating profile per day, with a 12-hour shutdown period at night.
However, instructions are included on how to set-up a second dispense period per day, if required.

Opening and closing hours set-up, beginning Sunday:

a. Carefully remove the lens from the front of the controller.
b. Press and hold the S button. Release after 2 seconds.
c. The display will change from Current Temperature to Current Set-point.
d. Continue to scroll through the menu, by pressing the S button for 2 seconds, then releasing, until the display shows "HOUr".
e. Press and release the ↵ button to enter the Opening/Closing hours mode.
f. The display begins with Sunday (displayed as sun). Note: the first 3 letters of each day is displayed digitally by LEDs, and is not particularly easy to decipher for some of the days.
g. To adjust the trading hours for Sunday, in 10 minute increments, follow this procedure....
h. Press and release the ↵ button again. A small LED flashes in the top left-hand corner of the display. This is the Opening Time.
i. Adjust the time up or down by using the **(+)** or **(-)** buttons.
j. When the desired time is displayed, press and release the ↵ button again.
k. The top left LED is extinguished, and a small LED flashes in the bottom left-hand corner of the display. This is the ***Closing Time.***
l. Adjust the time up or down by using the **(+)** or **(-)** buttons.
m. When the desired time is displayed, press and release the ↵ button again.
n. The bottom left LED is extinguished, and a small LED flashes in the top right-hand corner of the display. This is the Second Opening Time for Sunday. See opposite for Opening/Closing options...

Single opening/closing profile per day:
o. If no Second Opening Time is required, press and release the **(+)** button until the time scrolls past 24:00, and the next increment displayed is ***"OFF".***
p. Press and release the ↵ button twice to save the settings and move on to the next day **(Monday).**

Two opening and closing profiles per day:
q. If a Second Opening Time and Closing Time are required on the same day, this action should follow stage (n) above....
r. Adjust the Second Opening Time, using the **(+)** or **(-)** buttons.
s. When the desired time is displayed, press and release the ↵ button again.
t. The top right LED is extinguished and a small LED flashes in the bottom right-hand corner of the display. This is the Second ClosingTime.
u. Adjust the time by using the **(+)** or **(-)** buttons.
v. When the desired time is displayed, press and release the ↵ button again.
w. The display will now show Monday (displayed as mon).

Rest of the week...
x. Use the same procedure to adjust Opening and Closing hours for each day.
y. When the necessary adjustments have been made for all 7 days, press and release the S button. The display will show Python Refresh Mode (displayed as pytn). Ignore this mode.
z. Carefully replace the lens. The display will revert back to Current Temperature within 30 seconds.

No other adjustments should be necessary. Please contact Manitowoc Beverage Systems before changing any other parameters, as they may impact on energy savings, performance and ice build.

⚠ Warning

THIS UNIT IS UNSUITABLE FOR USE BY UNSUPERVISED CHILDREN, AGED OR INFIRM PERSONS.

⚠ Warning

THERE ARE NO END USER SERVICEABLE PARTS. ANY FAULT OR PROBLEM WITH THE EQUIPMENT MUST ONLY BE RECTIFIED BY A QUALIFIED SERVICE ENGINEER.

⚠ Warning

THIS EQUIPMENT NEEDS TO BREATHE. NEVER BLOCK-OFF THE AIRWAYS AROUND THE UNIT. DOING SO MAY SERIOUSLY AFFECT PERFORMANCE, AND MAY ULTIMATELY CAUSE TOTAL FAILURE. A MINIMUM AIR GAP ALL AROUND THE UNIT OF 75MM (3 INCHES) IS RECOMMENDED.

⚠ Warning

IT IS UNSAFE TO LIFT OR ATTEMPT TO MOVE THE COOLER DURING CLEANING OR AT ANY OTHER TIME WHILE THE UNIT IS OPERATING.

⚠ Warning

FAILURE TO KEEP THE CONDENSER CLEAN WILL RESULT IN THE UNIT MALFUNCTIONING AND OVERHEATING. IF THIS HAPPENS THE SAFETY CUT OUT ON THE COMPRESSOR WILL OPERATE, PREVENTING THE UNIT FROM WORKING UNTIL SUCH TIME AS IT HAS COOLED DOWN. IF THIS HAPPENS THE CONDENSER AND FILTER (IF FITTED) MUST BE CLEANED BEFORE USING AGAIN.

CARBON DIOXIDE WARNINGS WARNING!

⚠ Warning

THE CARBONATOR IS AN INTEGRAL PART OF THE UNIT AND IT SHOULD BE NOTED THAT THE CARBONATION PROCESS INVOLVES THE USE OF HIGH PRESSURES AND POTENTIALLY NOXIOUS GAS AND AS SUCH DUE CARE SHOULD BE TAKEN WHEN HANDLING, INSTALLING AND MAINTAINING THE EQUIPMENT WITH PARTICULAR REGARD TO THESE HAZARDS.

⚠ Warning

CARBON DIOXIDE LEAKS ARE POTENTIALLY FATAL IF CONCENTRATIONS RISE TO DANGEROUS LEVELS, IN VIEW OF THIS INSTALLATION SHOULD BE REGULARLY CHECKED FOR INTEGRITY AND THE GENERAL AREA OF INSTALLATION PROPERLY VENTILATED AT ALL TIMES.

⚠ Warning

CARBON DIOXIDE

1. ALWAYS	Connect the CO_2 or gas cylinder to a REDUCING VALVE.
2. NEVER	Try to connect cylinder directly to product container.
3. NEVER	Interconnect soft drinks, CO_2 or gas cylinder equipment with other equipment.
4. ALWAYS	Secure cylinder upright whilst in use.
5. ALWAYS	Keep cylinder away from heat.
6. NEVER	Drop or throw cylinders.
7. NEVER	Try to unscrew fittings from containers.
8. ALWAYS	Ventilate area after CO_2 leakage.

This information should be displayed in a position adjacent to the CO_2 supply cylinder at all times.

(FOR QUALIFIED ENGINEERS ONLY) ELECTRONIC CONTROLLER SETTINGS AND ADJUSTMENT INSTRUCTIONS.

DFX SERIES 1E CONTROLLER AND PROBE:
The controller is electronic, and is supplied with a temperature probe. The cut-out and cut-in temperatures are factory set at Manitowoc Beverage Systems, and should not require any adjustment.

They are as follows:

- **Cut-out -2.0°C**
- **Cut-in +0.5°C**

However, the controller has a built-in feature that allows each individual account operating profile to be set-up to optimise power usage. This feature conserves energy by powering down the cooler overnight, during the account closed period. There is a 7-day timer in the software, with factory default settings which shut the cooler down at 11:00pm and re-start the cooler at 11:00am. For accounts where this profile is inappropriate, the timer must be set-up to suit the individual account, in order to maximise energy savings. Please refer to the Dfx Series 1e Instruction Manual sent with the unit for guidance on setting up this feature.

Adjustment Guide:
Note: The Cut-out (1.) and Differential temperature (2.) are factory set at Manitowoc Beverage Systems. Any adjustment to these settings may affect the performance, energy saving characteristic and ice weight of the unit. It is not recommended that either of these parameters is adjusted without consultation with Manitowoc Beverage Systems Engineering Department.

1. To adjust the Cut-out temperature:

a. Carefully remove the lens from the front of the controller.
b. Press and hold the S button. Release after 2 seconds.
c. The display will change from Current temperature to Current set-point.
d. Adjust the temperature up or down using the **(+)** or **(-)** button.
e. To store the new set-point, scroll through all menu options to the screen which displays _.__ The controller will then revert to the current temperature.
f. The display will then show the differential temperature. If no differential adjustment is necessary, replace the lens. The unit will revert to the Current Temperature after 30 seconds.

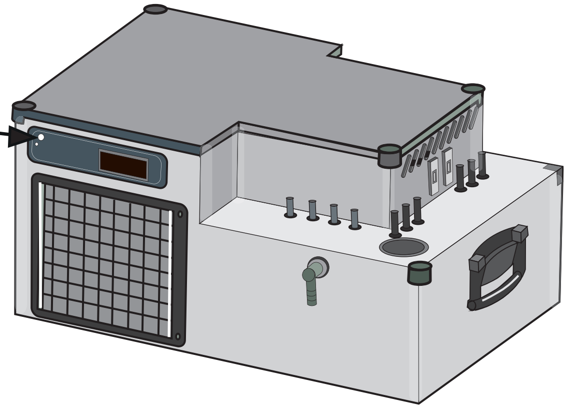
2. To adjust the differential (cut-in) temperature:

a. Follow steps (a) and (b) above.
b. The display will change from Current Temperature to Current Set-point.
c. Press and hold the S button again. Release after 2 seconds.
d. The display will change from current set-point to current differential.
e. Adjust the temperature up or down using the **(+)** or **(-)** button.
f. To store the new set-point, scroll through all menu options to the screen which displays _.__ The controller will then revert to the current temperature.
g. The display will then show the Diagnostic Mode, or 9999. Ignore this mode. Replace the lens. The unit will revert to the Current Temperature after 30 seconds.

Example: If the cut-out temperature is set at -2.0°C, the differential must be set to 2.5°C to cause the unit to cut-in at 0.5°C.

BI-LEVEL CARBONATOR CONTROLLER TIME OUT SAFETY FEATURE

The carbonator has a time out feature that will shut the carbonator pump off and the soda recirculation pump off if the bowl does not fill in 1 min 30 seconds. This is to protect both pumps in the event of the water starvation. If the time out feature operates, the small LED on the front panel will flash and the carbonator pump and soda recirc pump will shut off. To reset, ensure the water supply has been restored, turn off the power to the unit for 5 seconds and then switch back on. If the bowl still fails to fill, then the fault must be investigated and rectified before the unit can be operated.



END USER OPERATIONAL INFORMATION AND WEEKLY BASIC INSPECTION CHECKS

- This equipment must be installed and maintained by a qualified Engineer. No end user adjustments or alterations are necessary.
- Once installed, there is no user interface required, other than the basic end user weekly inspection checks listed below.
- There is no end-user training required on this unit other than understanding the weekly basic checks for safe unit operation.
- No operator training is required, as this quipment is fully automated and contains no user interfaced functions. No operator adjustment is necessary.

WEEKLY BASIC INSPECTION CHECKS BY THE END USER

It is not recommended that the End User makes any adjustments or carries out any maintenance other than:

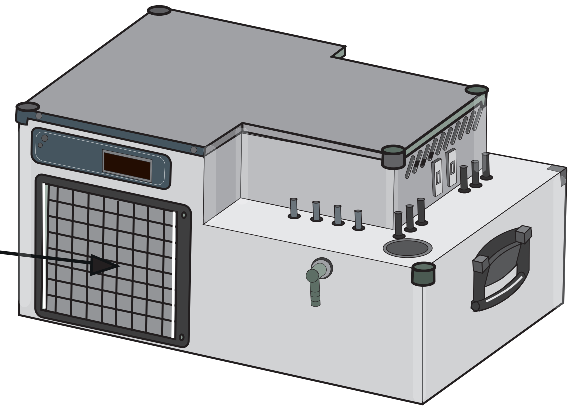
- Visually Check the condition of the mains lead and plug.
- Visually check the unit and its pipe work for evidence of leaks.
- Check that the condenser grill and vents are not choked or obscured.
- Clean the mesh filter (if fitted) and condenser with a soft brush or vacuum cleaner to remove any dust/debris.

INSTRUCTIONS FOR CLEANING THE CONDENSER & MESH FILTER (WHERE FITTED)

The condenser and filter (if fitted) at the front of the unit should be cleaned weekly using a soft brush or vacuum cleaner to remove any extraneous debris.

To remove the mesh filter (if fitted) insert a flat blade screwdriver or thin coin in the slot of the plastic fastener that secures the metal grill at the front. Turn the fastener a 1\4 turn to release the grille. Replacement is the reverse of removal.

IMPORTANT – Always replace the mesh filter and securing grille after cleaning.



* FOR QUALIFIED ENGINEERS ONLY

INSTALLATION AND CONNECTION INSTRUCTIONS FOR PUTTING THIS EQUIPMENT TO USE.

GENERAL

- Unpack the unit from its transportation packing and visually check for any signs of damage.
- Place the unit in a convenient location in the cellar or room where it is to be sited.
- Always site the cooler on a dry, level surface to avoid e.g. spillage and any unnecessary noise or vibration. If mounting the unit off the floor, ensure that:-

o The surface is clean, flat and dry,
o The surface is of sufficient size to support the unit without it overhanging,
o The surface can support the weight of the full, operational unit,
o The unit, when mounted, is stable and cannot slip or be pushed off the surface. If necessary, secure the unit and/or mounting surface to a solid fixing point such as an outside wall.

- Do not site unit beside heat sources, e.g. dishwashers.
- Make sure that a mains electricity supply is within 2 meters and in an area allowing free circulation of air.
- The unit should be fitted with a correctly fused and wired 13 amp plug fitted with a 10 amp fuse or alternatively a standard Euro Plug to IEC83:1975.

CONNECTING THE UNIT TO THE WATER, CO_2 AND PYTHON AND COMMISIONING

1. Connect the mains water feed to the connection at the rear of the unit labelled WATER IN. It is recommended that the maximum \ minimum water pressure to the unit is **50 \ 20 psi.**
   An inline water pressure regulator set at **50 psi** and a stop tap or other water shut off device should also be installed in line to shut off the water feed to the unit should an internal leak occur within the unit**. (Do not turn water on yet)**
2. Connect the CO2 feed to the connection labelled CO2 IN at the rear of the unit. It is recommended that the maximum \ minimum CO2 pressure supply to the unit is **80 \ 75 psi.** At pressures below this, it is possible that the unit will not carbonate the water correctly resulting in flat drinks. **(Do not turn the CO2 on yet)**
3. Remove the bath filler cap and fill the unit with clean cold water until the top evaporator coil is covered.
4. Dose the water bath (IF REQUIRED) with Aquacid CS cooler water control agent in accordance with instructions supplied on the bottle, taking note of health & safety instructions.
5. Connect the syrup inlet lines to the syrup coils labelled coils **1- 4.**
6. Connect the appropriate lines within the python to the cooler, identifying from the labels on the cooler, the soda recirc flow and return, the other end of the syrup coils 1-4 and the still water coil (if required). Insulate all connections made using appropriately sized tube insulation.
7. Connect the other end of the python to the dispense head.

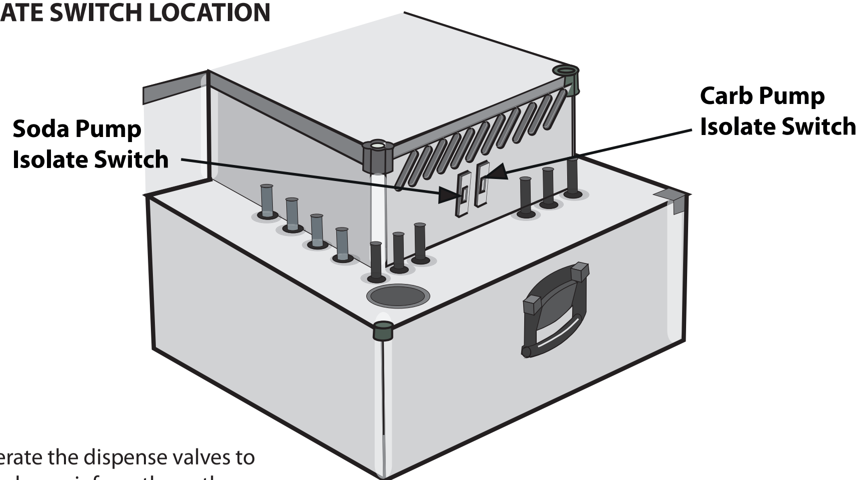
Note: Ensure that the recirc run is not excessive, and that the python is routed away from external heat sources. Maximum lift for the recirc pump is 6m. Maximum recommended run is 12 metres.

8. When all connections have been made, turn on the CO2 supply and allow the system to fill with CO2. Lift the relief valve on the top of the integral carbonator bowl which is found underneath the coil deck lid to expel any air from the bowl.

Note: Any unused syrup coils should remain capped both for personal protection, and to prevent the ingress of debris.

Note: If the fridge unit is to be operated to make the ice-bank whilst the installation is being completed, isolate the carbonator pump and the soda recirc pump by switching off the two switches on the end panel. If the carbonator pump switch is left in the off position, the time out feature will operate after 1min 30 secs, the LED will flash and the soda recirc pump will cut out. To reset, the power to the unit must be switched off for 5 seconds and then back on .

CARB PUMP AND SODA PUMP ISOLATE SWITCH LOCATION



9. Operate the dispense valves to expel any air from the python.
10. Check the installation for any gas leaks. Rectify if any are found before proceeding.
11. Ensure the soda pump switch is in the off position.
12. Turn on the water supply and again check for leaks.
13. Ensure the carbonator pump switch is in the ON position.
14. Plug the unit into the mains supply and switch on.

Note: There is approximately a one minute delay from switching the cooler on, to the compressor starting.

15. Allow the carbonator to fill and then by opening the dispense valves, bleed the gas pockets from the system until the python is full of carbonated water.
16. Switch the soda recirc pump on and allow the system to make its ice bank and chill the python down.
17. Connect the syrup supply and brix the system when down to temperature.